ENGR 1451/2451: Exploratory Data Science

Comparing Many Means with ANOVA

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Today: Beyond Comparing Two Means

Review: inference for two groups

A two-sample t-test asks whether **two population means** differ, accounting for sampling variability.

New question

Suppose we have **three or more groups**. How do we test whether their means are all equal without running many separate tests?

Road map

Visualize the groups $\rightarrow$ state hypotheses $\rightarrow$ compare **between- and within-group variability** $\rightarrow$ compute F and p $\rightarrow$ consider pairwise comparisons.

A Quick Review: Two Means Versus Many

Two independent groups

$H_0 : \mu_1 = \mu_2$

$$T = \frac{\bar{y}_1 - \bar{y}_2}{SE(\bar{y}_1 - \bar{y}_2)}$$

A large $|T|$ is evidence against equality.

Three or more groups

$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

A large F is evidence against equality.

ANOVA means **AN**alysis **O**f **VA**riance, even though the hypotheses concern means.

Example: Aldrin in Tennessee's Wolf River

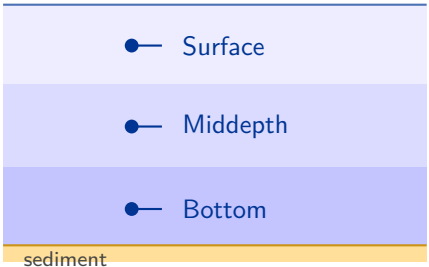
Environmental question

Aldrin is an insecticide associated with historical waste dumping near the Wolf River. Does its **mean concentration** differ with sampling depth?

Variables

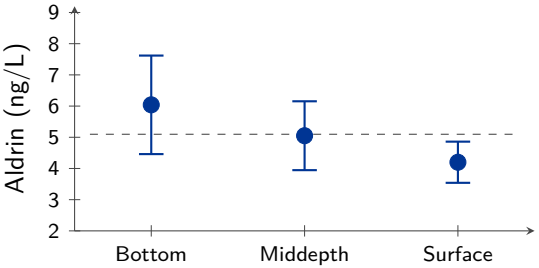
Response: aldrin concentration (ng/L).

Group: surface, middepth, bottom.



Schematic of the three sampling depths (not to scale).

Start with the Data, Not the p-Value



Points = sample means; error bars = *one sample SD*, not confidence intervals. Dashed = grand mean.

Summary

Depth	n	Mean	SD
Bottom	10	6.04	1.58
Middepth	10	5.05	1.10
Surface	10	4.20	0.66
Overall	30	5.10	1.37

Question: Are the observed mean differences large relative to the within-depth variation?

Discussion: What Exactly Does ANOVA Test?

Three depths: bottom (B), middepth (M), and surface (S).

Select the correct hypotheses

1. $H_0 : \mu_B = \mu_M = \mu_S$; $H_A : \mu_B \neq \mu_M \neq \mu_S$.
2. $H_0 : \mu_B = \mu_M = \mu_S$; H_A : at least one population mean differs.
3. $H_0 : \mu_B = \mu_M = \mu_S = 0$; H_A : at least one mean differs.

Follow-up: Does the alternative require *every pair* of means to differ?

Why Not Just Run Three t-Tests?

All three pairwise questions

1. Bottom versus middepth
2. Bottom versus surface
3. Middepth versus surface

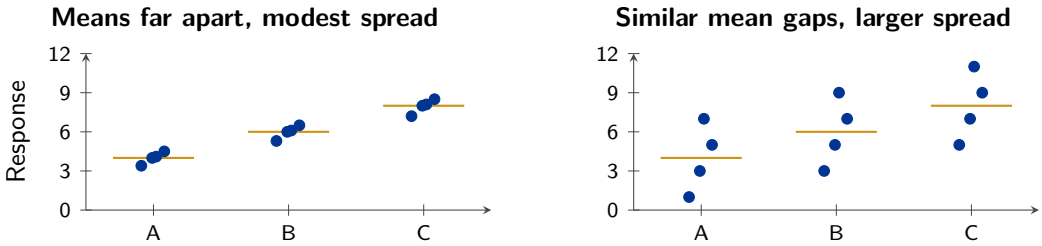
The multiple-testing problem

Each test at $\alpha = 0.05$ can falsely reject a true null.
Running more tests increases the chance of at least one false positive.

ANOVA's role

Start with one global test of equal means. If needed, investigate which groups differ with a multiple-comparison procedure.

The Big Idea: Signal Relative to Noise



Question: Which pattern should give stronger evidence of unequal means, and why?

ANOVA Compares Two Sources of Variation

Between groups: signal

How far is each group mean from the grand mean $\bar{y}$?
Larger separation $\Rightarrow$ larger between-group variability.

Within groups: noise

How far is each observation from its own group's mean?
Larger scatter $\Rightarrow$ larger within-group variability.

A large F means the between-group signal is large relative to the within-group noise.

Between-Group Sum of Squares: SSG

Compare each group mean to the grand mean

$$SSG = \sum_{j=1}^{k} n_j(\bar{y}_j - \bar{y})^2$$

- k: number of groups.
- n_j: observations in group j.
- $\bar{y}_j$: mean of group j.
- $\bar{y}$: mean of all observations.

Wolf River: $k = 3$, $n_j = 10$

$$\begin{aligned} SSG &\approx 10(6.04 - 5.097)^2 \\ &\quad + 10(5.05 - 5.097)^2 \\ &\quad + 10(4.20 - 5.097)^2 \end{aligned}$$

$$\boxed{SSG \approx 16.961}$$

Within-Group Sum of Squares: SSE

Compare each value to its own group mean

$$SSE = \sum_{j=1}^{k} \sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2$$

This is the variation not explained by group membership.

Wolf River

$$SSE = 37.329$$

Each of the 30 measurements has a residual relative to its depth's sample mean.

Connection to regression: SSE again means the sum of squared residuals.

Total Variation = Between + Within

The same decomposition used in regression

$$\underbrace{SST}_{\text{Total}} = \underbrace{SSG}_{\text{Between groups}} + \underbrace{SSE}_{\text{Within groups}}$$

$$54.290 \approx 16.961 + 37.329$$

Earlier: simple regression

$SST = SSR + SSE$

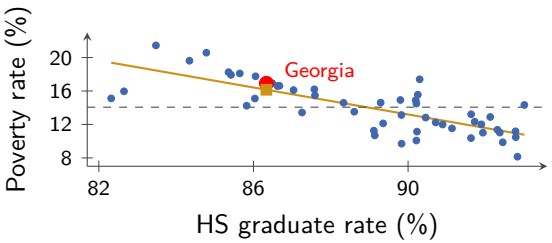
SSR: variation accounted for by the fitted line.

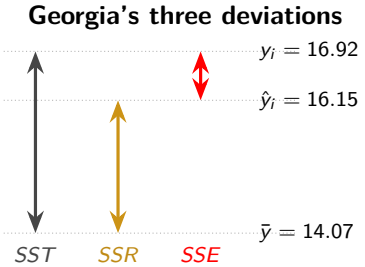
Now: one-way ANOVA

$SST = SSG + SSE$

SSG: variation accounted for by the group means.

Visualizing *SST*, *SSR*, and *SSE* in Regression



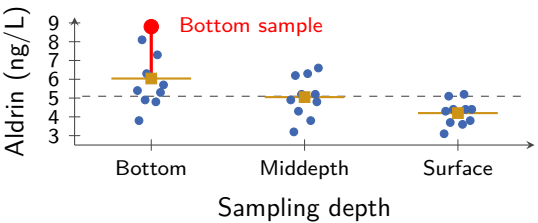


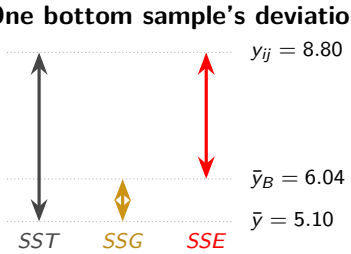
Blue: observations; gold: fitted line; dashed: overall mean.

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{SST} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{SSE}$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \approx 0.599$$

Visualizing *SST*, *SSG*, and *SSE* in One-Way ANOVA





Measured values from `aldrin.csv`; gold: group means; dashed: grand mean.

$$\underbrace{\sum_j \sum_i (y_{ij} - \bar{y})^2}_{SST} = \underbrace{\sum_j n_j (\bar{y}_j - \bar{y})^2}_{SSG} + \underbrace{\sum_j \sum_i (y_{ij} - \bar{y}_j)^2}_{SSE}$$

$$R^2 = \frac{SSG}{SST} = 1 - \frac{SSE}{SST} = \frac{16.961}{54.290} \approx 0.312$$

Where Do the Degrees of Freedom Come From?

General formulas

$$df_G = k - 1 \qquad \text{(groups)}$$

$$df_E = n - k \qquad \text{(error)}$$

$$df_T = n - 1 \qquad \text{(total)}$$

Check: $df_T = df_G + df_E$; here $29 = 2 + 27$.

Our three depths

$k = 3$ groups, $n = 30$ total samples.

$$df_G = 3 - 1 = 2$$

$$df_E = 30 - 3 = 27$$

$$df_T = 30 - 1 = 29$$

Turn Sums of Squares into Mean Squares

Between-group mean square

$$\boxed{MSG = \frac{SSG}{k-1}}$$

$$MSG = 16.961/2 \approx 8.480$$

Within-group mean square

$$\boxed{MSE = \frac{SSE}{n-k}}$$

$$MSE = 37.329/27 \approx 1.383$$

Why divide by degrees of freedom?

MSE estimates common within-group variance σ^2. Under H_0, MSG also estimates σ^2.

Do not confuse ANOVA MSE with the earlier RMSE

$$\underbrace{\mathrm{RMSE} = \sqrt{SSE/n}}_{\text{in-sample error (response units)}} \qquad \underbrace{\sqrt{MSE} = \sqrt{SSE/(n-k)}}_{\text{residual SD estimate (response units)}}$$

The F-Statistic: Between Divided by Within

F-test for equality of group means

$$\boxed{F = \frac{MSG}{MSE} = \frac{SSG/(k-1)}{SSE/(n-k)}}$$

$$F = \frac{16.961/2}{37.329/27} \approx \boxed{6.134}$$

If H_0 is true: both mean squares estimate the same variance, so an F around 1 is not surprising.

If group means differ: the between-group mean square tends to be larger and F tends to increase.

Read the ANOVA Table

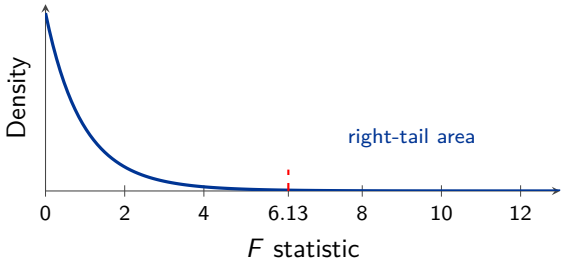
Source	df	Sum Sq	Mean Sq	F	p
Depth (groups)	2	16.961	8.480	**6.134**	**0.00637**
Error (within)	27	37.329	1.383		
Total	29	54.290			

Reading the columns

Sum Sq: total squared deviations. **Mean Sq**: Sum Sq / df.

F: MSG / MSE. p: upper-tail probability under equal population means.

Where Does the ANOVA p-Value Come From?



Reference distribution: F with $(2, 27)$ degrees of freedom.

Under H_0

$$F \sim F_{2,27}$$

$$p = P(F \geq 6.134) \approx 0.00637$$

Interpretation: If all three population means were equal, this large an F would be unusual.

Discussion: What Can We Conclude?

For the three depths, $F(2, 27) \approx 6.134$ and $p \approx 0.00637$.

Which statement is supported at $\alpha = 0.05$?

1. All three population means are different from one another.
2. At least one population mean differs from another, if the model's assumptions hold.
3. Aldrin is necessarily higher at the bottom because of depth.
4. A p-value of 0.00637 means there is a 0.637% chance H_0 is true.

Conditions for One-Way ANOVA

1. Independent observations

Samples should be independent **within and between** groups. Check the sampling design; observations paired by location or time are not independent.

2. Approximately normal distribution within each group

Check group-level histograms or normal Q–Q plots, especially when sample sizes are small or outliers are present.

3. Approximately equal group variances

Compare group SDs or boxplot spreads. This matters particularly with unequal sample sizes.

Do the Wolf River Data Meet the Conditions?

What can we check here?

Depth	n	SD (ng/L)
Bottom	10	1.58
Middepth	10	1.10
Surface	10	0.66

Largest/smallest SD ≈ 2.39.

What requires more information?

- Were samples independent or collected at the *same sites* across depths?
- Are there strong departures from normality in individual groups?
- Are spreads sufficiently similar for the equal-variance model?

Question: Which assumption concerns you most? What would you request from the sampling team?

Python: ANOVA from an Actual DataFrame

Using statsmodels with the 30 observations in `aldrin.csv`

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

import pandas as pd
# Read aldrin.csv from the data/ folder:
aldrin_data = pd.read_csv("data/aldrin.csv")
model = smf.ols("aldrin ~ C(depth)", data=aldrin_data).fit()
print(sm.stats.anova_lm(model, typ=2))
```

`C(depth)` instructs Python to treat depth as a **categorical predictor**, not a numeric slope.

`anova_lm` gives the group and error sums of squares, degrees of freedom, F, and p.

Which Means Differ? Multiple Comparisons

The ANOVA conclusion is global

$p < 0.05$ tells us **not all group means are equal**. It does *not* identify the differing pairs.

Three groups: three pairs

1. Bottom–middepth
2. Bottom–surface
3. Middepth–surface

Bonferroni adjustment

$$K = \frac{k(k-1)}{2} = 3$$

$$\boxed{\alpha^* = \frac{0.05}{3} = 0.0167}$$

Worked Pairwise Comparison: Bottom vs Surface

Difference of sample means

$$\bar{y}_B - \bar{y}_S = 6.04 - 4.20 = 1.84 \text{ ng/L}$$

Pooled standard error from ANOVA

$$SE = \sqrt{MSE\left(\frac{1}{n_B} + \frac{1}{n_S}\right)}$$
$$\approx \sqrt{1.383(1/10 + 1/10)} = 0.526$$

Two-sided pairwise t-test

$$t = \frac{1.84}{0.526} \approx 3.50$$

$df = 27$; $p \approx 0.00164$. **Bonferroni:** $0.00164 < 0.0167$.

Conclusion: Under the equal-variance assumptions, bottom and surface have evidence of different population means.

Discussion: Which Pairwise Differences Survive?

For pooled comparisons computed from `aldrin.csv`, use $\alpha^* = 0.0167$.

Comparison	Mean diff.	t_{27}	Two-sided p	Decision
Bottom – middepth	0.99	1.88	0.0706	Not significant
Bottom – surface	1.84	3.50	0.00164	**Significant**
Middepth – surface	0.85	1.62	0.1176	Not significant

Question: Does a nonsignificant result prove the corresponding population means are equal?

Connection Back to Regression and Two-Sample Tests

One-way ANOVA as regression

A categorical predictor can be represented by group indicators; the fitted response is each group's mean.

$$SST = SSG + SSE$$

When there are only two groups

With the **same pooled equal-variance assumption**, ANOVA and the two-sided pooled t-test are equivalent:

$$\boxed{F_{1,n-2} = t_{n-2}^2}$$

Same p-value.

For three or more groups: the global F-test assesses the collection of means at once.

Summary: Which Statistical Test Should You Use?

What are you testing?	Statistical test
Numerical response: means	
One mean versus a target; population σ known	One-sample z-test
One mean versus a target; σ unknown	One-sample t-test
Two independent group means	Two-sample t-test (Welch)
Before/after measurements on the same units	Paired t-test
Three or more independent group means	**One-way ANOVA (F-test)**
Categorical response: counts	
Observed counts versus specified category probabilities	χ^2 goodness-of-fit
Association between two categorical variables	χ^2 independence
Compare category distributions from different groups	χ^2 homogeneity

Takeaways: Comparing Many Means

1. **Question:** $H_0 : \mu_1 = \cdots = \mu_k$; the alternative is that *at least one mean differs*.
2. **Decomposition:** $SST = SSG + SSE$ separates between- and within-group squared variation.
3. **Test:** $F = MSG/MSE$, with degrees of freedom $(k-1, n-k)$; a large F gives a small upper-tail p-value.
4. **Validity:** independence, within-group shape, and similar variances must be evaluated.
5. **After a significant ANOVA:** use adjusted pairwise comparisons to determine *which groups* differ.